

**Ahmedabad
University**

Simple Reaction Time Task Experiment

PSY310: Lab in Psychology

Lab Report

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GitHub Link: <https://github.com/aayushi111525/Simple-Reaction-Time-Task>

Introduction:

Reaction time basically measures how fast a person reacts to something. Donders (1868) was one of the people who studied this. In this experiment, simple reaction time task was used. This means that there will be only single stimulus and one response to it. The participant won't have any option to choose from, they just had to respond as fast as they could. Because this is a simple reaction time task, it will just measure on how fast they are moving their hand and it's not about decision making or any other aspects.

Method:

The experiment was a simple reaction time task. The task had two parts i.e. a fixation and then a target. At the start of each trial, a circle which is the fixation appeared in the middle of the screen. The participant would look at the centre of the screen and get ready. Following the fixation, the target was displayed which was a square.

The task is for the participant to press 'Y' key as soon as they see the square (i.e. the target). The time between the square appearing and the participant pressing 'Y' was recorded which would be their reaction time. The time was recorded in milliseconds.

The participant was given all the instructions before the experiment and was also told to keep their finger near 'Y' key so that they could respond quickly. The task had 50 trials, and the reaction time for each trial was recorded and saved as an excel file. This data was then used to calculate the average reaction time, median and standard deviation.

Participants:

There was only 1 participant on whom the experiment was conducted and he was a 20 year old male. He had normal vision and no health issues which could affect the reaction time. Verbal consent was taken before the experiment. They agreed to take part in the experiment.

Materials and Stimuli:

The experiment was done using PsychoPy on a windows run laptop. The participant has to respond to the stimuli [square] by pressing 'Y' on the keyboard.

Two shapes were used:

- Fixation - it was a circle, shown first to get the participant ready and keep their eyes fixated in the centre of the screen.
- Target - it was a square which was shown after the fixation. When the square appeared, the participant had to press 'Y' as fast as they could.

Image 1:

The fixation[circle] is presented at the start of each trial. This was shown on the centre of the screen to keep participant's attention at the centre before the target stimulus [square] appeared.

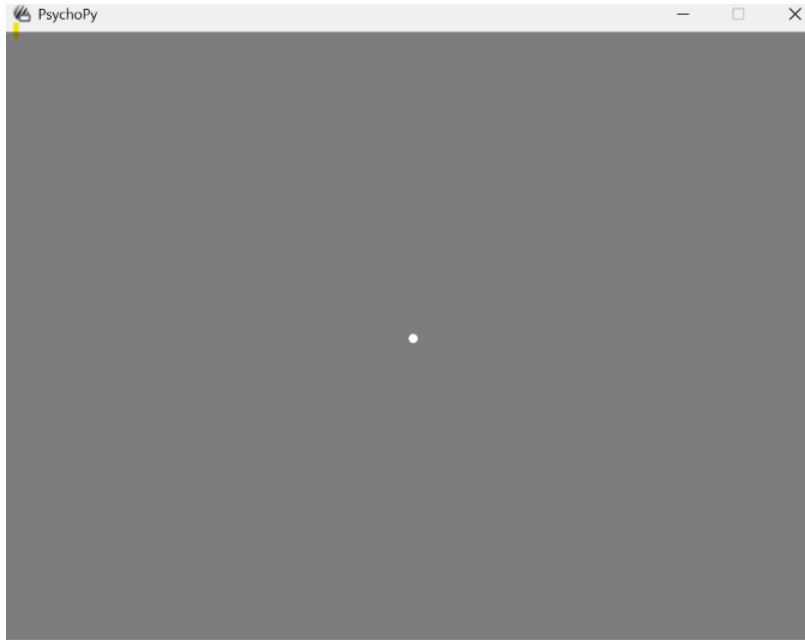
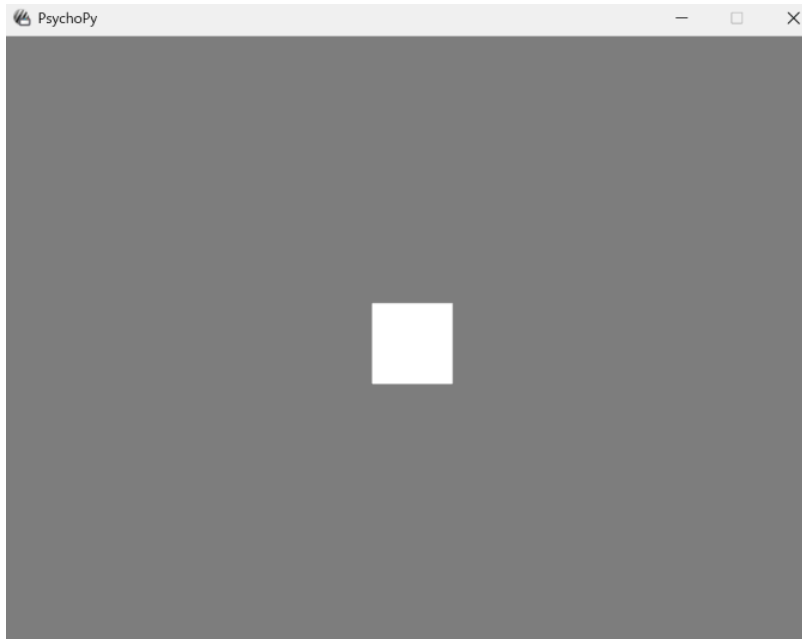
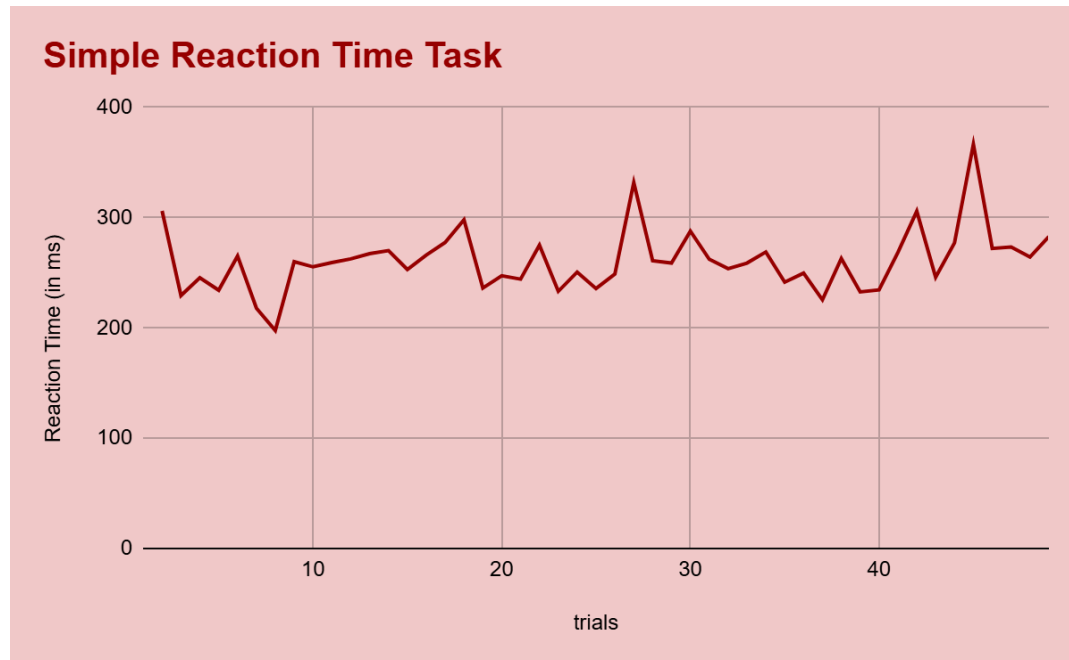


Image 2:

The target [square] was shown 0.5 seconds after the fixation and the participant had to press 'Y' as fast as possible.



Results:



According to the data, across all the 50 trials, the participant's reaction time to the square was on average 261.78 ms. The standard deviation was 29.12 ms. The median was 259.65 ms. There were variations between the trials with trial 7 being the fastest (197.65 ms) and the slowest one being 366.63 ms which was trial 44. Overall, all reaction times were falling in the normal range, with a few trials scattered in the middle and end.

Discussion:

The average reaction time was around 262 ms which is actually normal for simple reaction time task which says that the average reaction time lies between 200-300 ms. This suggests that the participant's reaction time was normal. The increase in the reaction time after half trials could be because of the participant losing focus, or being tired and not getting better with practice, because if that were the case, then the reaction time would've been faster.

Limitations:

This experiment had only 1 participant, so whatever the results were, it cannot be generalized to other people. The reaction time also depends on a lot of factors other than vision, like age, tiredness, how easily they can get distracted etc. In addition to that things like noise or distraction in the room wasn't controlled. In future, we could use more participants to get more valid results.

References:

looti, M. (2026, August 2). Mental Chronometry: Measuring the Speed of Human Thought. Encyclopedia of psychology. <https://encyclopedia.arabpsychology.com/donderss-law/>